Good software to learn

It's a good idea at some point in your uni maths journey to learn some software to help you do computations. This will enable you to check your answers to certain calculations or key steps. Some main examples include calculating derivatives and integrals, finding solutions to DE's and calculating vector norms (when we get to those concepts).

These pieces of software are non-random. They are 'computer algebra systems' (CAS's) and will give you the same answer each time you re-run a calculation. I make this point as your favorite LLM might know the methods to approach a problem, but they won't always get the numbers right (and the numbers are often very important).

My favourite CAS's are:

1. Wolfram Alpha. This is the most user friendly and runs in the browser. It's a bit clumsy to use but it works. It's free.

2. Mathematica. This is also made by Wolfram, but runs locally on your machine (i.e. you download it). It's not too difficult to download and get set up, and I recommend eventually using this over Wolfram alpha in the browser. It takes a bit longer to learn than using Wolfram Alpha and can feel like programming but is worth it and first time programmers shouldn't be scared.

   This software runs very fast and is very extensible. You can solve differential equations, make plots, anything we need in calculus. It's very good software and is freely available when you download/login with your uni credentials.

3. Sage Math. This software is similar to mathematica but open source, and hence is free. It's good and popular software but isn't as unanimous as mathematica.

4. Matlab. Matlab is used in the class Linear Algebra. It is decent software but not something I go back to much. It's very powerful and used a lot in engineering related work, but is not the mathematicians favorite (that is mathemtica or sage... in my opinion). I believe you can do most of the calculus stuff we do here with it, and learning it will help you if you take linear algebra.

5. There are some python libraries which you can use, e.g. scipy and sympy, both of which I like and use, but you need a place to run python.

Overall I'd recommend you use Mathematica, Matlab or Sage, but you will need to dedicate a day or two to getting them setup and learning how you do calculations with them. This is worth it and you will use this software throughout your maths studies.

While I'm on this note, if you haven't already, learn to use something like Desmos graphing calculators to make graphs and help you play around with equations. Desmos is often a bit quicker and easier to play around with multiple plots than Mathematica, and it runs in the browser.